



DIGITAL FORENSICS IN NETWORKS-3332-SEC-3

Chapter 1: Network Security Review

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Why Network Security Matters to Forensics



Network security encompasses the policies, technologies, and strategies that protect information assets from cyberattacks.



Traditional mechanisms — firewalls, access controls — remain important, but modern adversaries have adapted to bypass them.



Security tools do more than prevent attacks: they generate the logs, alerts, and packet captures that become forensic evidence.

Question?

If a firewall blocks every attack it sees, what forensic value does it still provide the investigator?

Answer: even blocked attempts are logged — those logs establish attacker intent, timing, and technique.

Key Security Layers

Traditional mechanisms remain important even as adversaries adapt to bypass them.



Perimeter Security

Firewalls, IDS/IPS systems, VPN gateways.



Host-Based Security

Antivirus, EDR (Endpoint Detection and Response), host firewalls.



Application Security

Secure coding practices, input validation, patch management.



Monitoring & Logging

Centralized logging with SIEM platforms (e.g., Splunk, ELK Stack).

Deep Dive: Perimeter & Host-Based Security



Perimeter Security

Firewall

Filters traffic by rule (IP, port, protocol); logs allow/deny decisions.

IDS/IPS

Detects or blocks known attack signatures and anomalies in transit.

VPN Gateway

Encrypts remote access; connection logs show who, when, from where.



Host-Based Security

Antivirus

Signature/heuristic scanning; quarantine events are evidentiary.

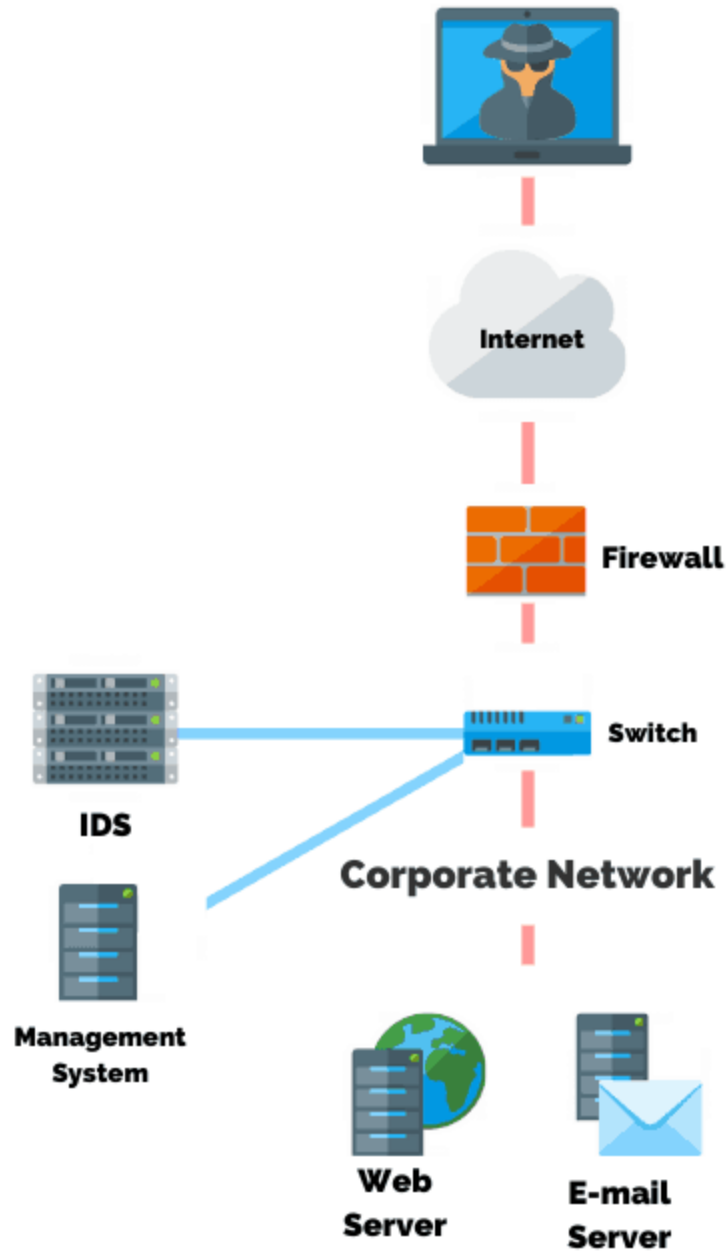
EDR

Endpoint Detection & Response — records process trees, file/registry changes.

Host Firewall

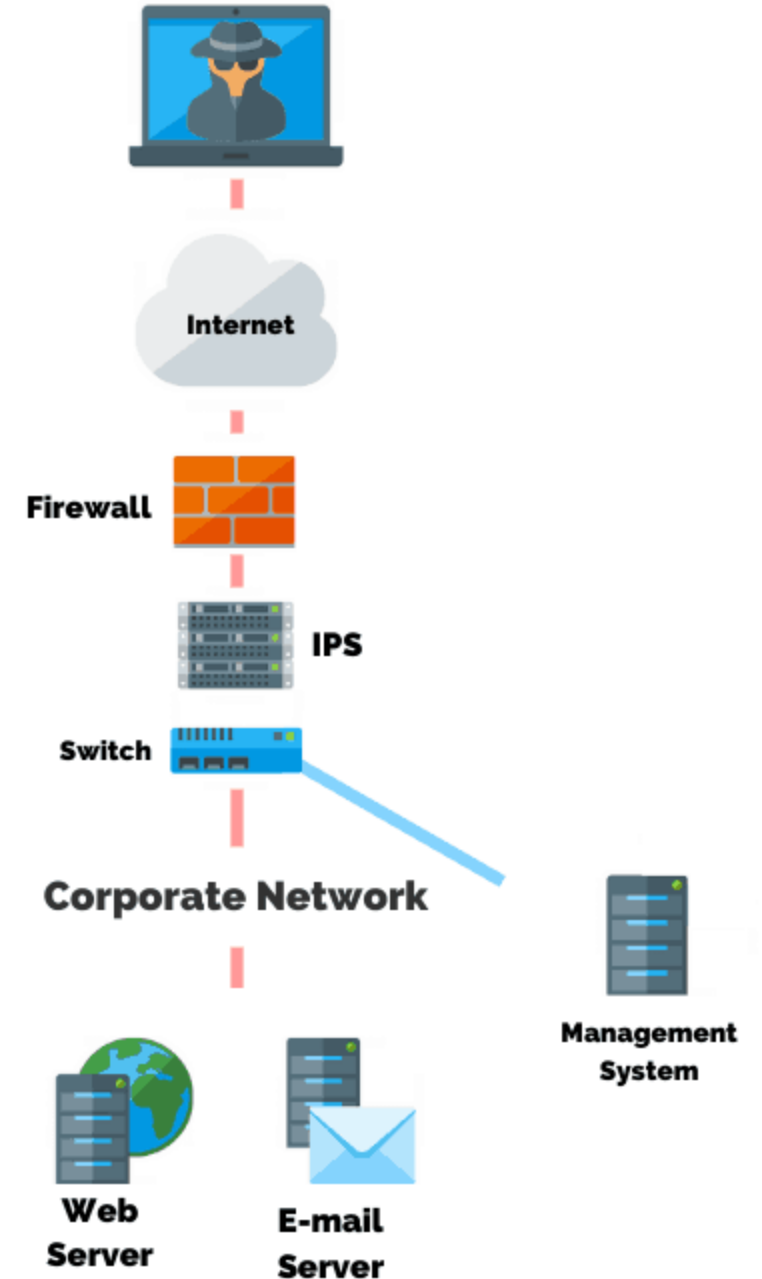
Local traffic rules; complements perimeter defenses per device.

Intrusion Detection System (IDS)



VS

Intrusion Prevention System (IPS)



Deep Dive: Application Security & Monitoring



Application Security

Secure Coding

Reduces exploitable flaws (injection, buffer overflow) at the source.

Input Validation

Rejects malformed/malicious input before it reaches business logic.

Patch Management

Closes known vulnerabilities — the WannaCry case (Part 4) shows the cost of delay.



Monitoring & Logging

SIEM (Splunk, ELK)

Centralizes and correlates logs across the whole environment.

Alerting

Flags anomalous patterns for analyst triage in near real time.

Blind Spots

Weak/incomplete logging is exactly where attackers hide — the central forensic risk.

Input Validation

cymulate



User Input

Name, Email,
Comment, etc.



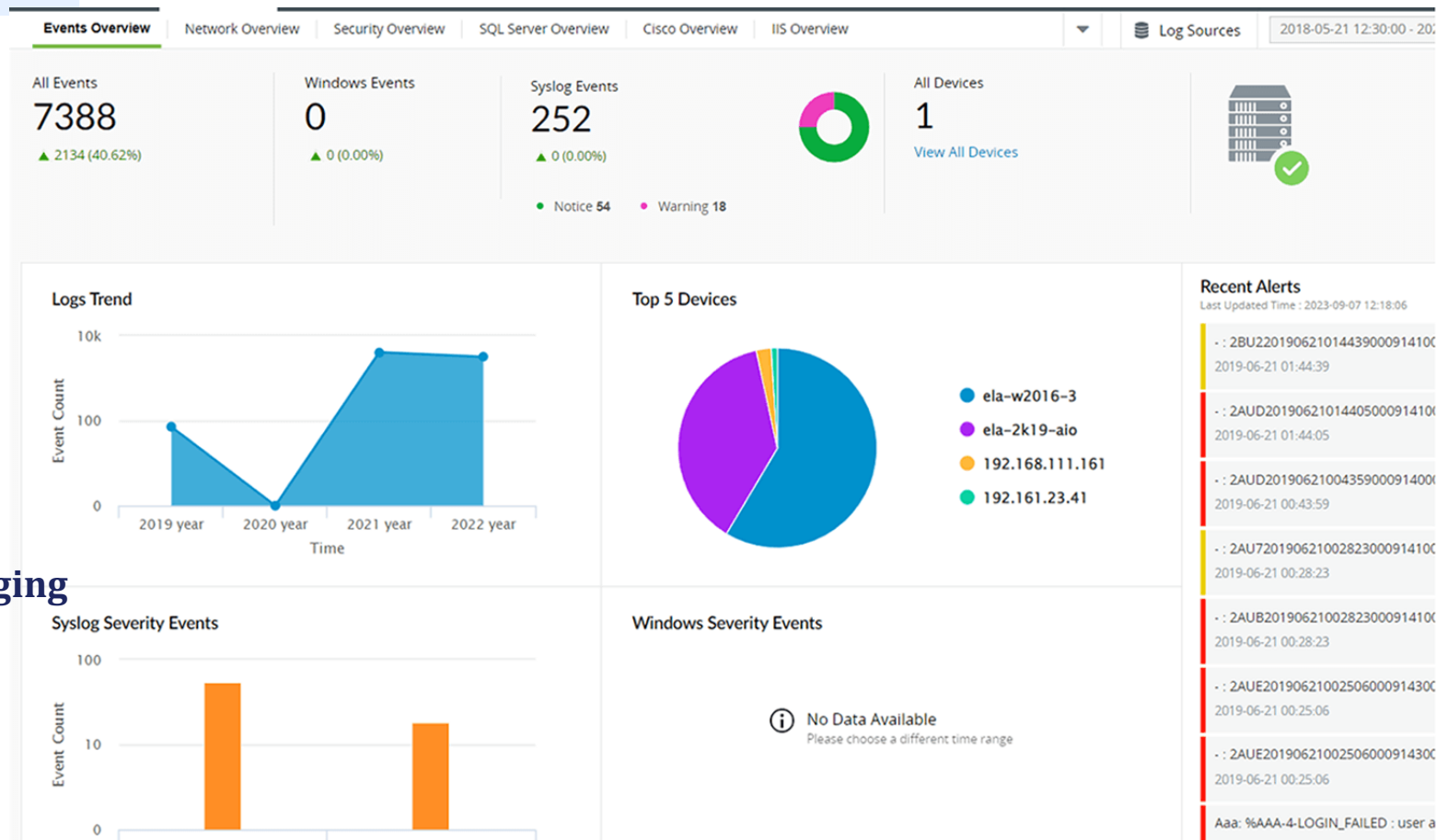
Validation Check

Correct format?
Correct type?
Acceptable range?



Accept or Reject

Monitoring & Logging



Forensic Relevance



Security tools generate logs, alerts, and packet captures that form the basis of forensic evidence.



Attackers often leave traces in logs such as failed login attempts, anomalous network flows, or unusual DNS queries.



Weak or incomplete logging results in "blind spots" that adversaries exploit.

“

Attackers often leave traces in logs such as failed logins, anomalous flows, or unusual DNS queries.

— Davidoff (2013)

Textbook cross-reference: Ch. 1, on log-based attacker traces.



PART 2

Digital Evidence Concepts

Digital data is fragile and easily altered — forensic principles must be followed

What Counts as Digital Evidence?



Digital evidence refers to information that can establish or refute facts in an investigation. Unlike physical evidence, digital data is fragile and easily altered, so forensic principles must be followed.

Three core principles govern how we handle it:



Order of Volatility

Collect the most volatile evidence first.



Integrity & Hashing

Verify with MD5/SHA-256; a mismatch invalidates evidence.



Chain of Custody

Document every handler; courts require an unbroken chain.

Order of Volatility (OoV)

Digital evidence is fragile — investigators must collect the most volatile evidence first.



Class question: If a compromised server must stay online for business reasons, which evidence source should you image first — and why?

Integrity & Chain of Custody



Integrity & Hashing

- Evidence must be verified with cryptographic hashes (MD5, SHA-256) to prevent tampering.
- A mismatch in hash values invalidates the evidence.



Chain of Custody

- Every person who handled the evidence must be documented.
- Courts only accept evidence with a provable, unbroken chain of custody.

Challenges in Network Forensics



Encryption

SSL/TLS hides attacker payloads.



Data Volume

High-speed networks generate terabytes of traffic, requiring selective capture.



Evasion Techniques

Attackers fragment packets, use steganography, or spoof traffic.

“Even a single captured packet may contain the evidence needed to prove or disprove a hypothesis.” — Davidoff

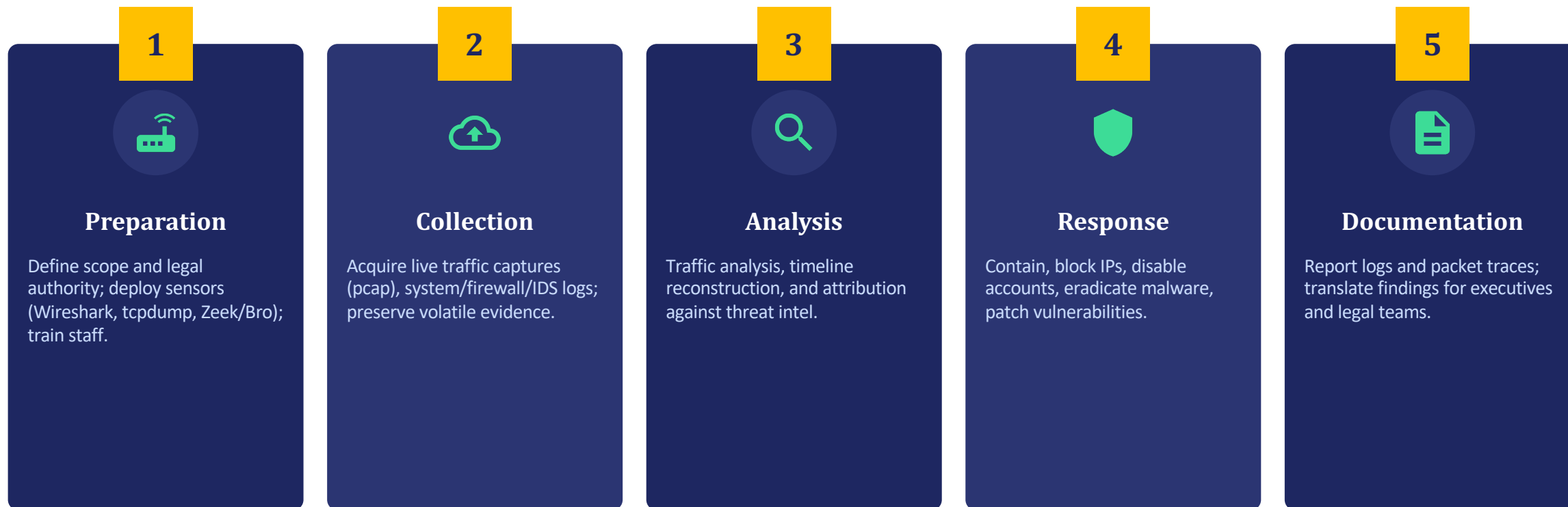


PART 3

Practical Investigation Strategies

Effective network forensic investigation follows a structured methodology.

A Structured Investigation Methodology



Note the sequence: Preparation and Collection must happen before analysis is even possible — this is why OoV (Part 2) matters operationally, not just academically.



PART 4

Real-World Cases

Technical focus: how network forensics actually exposed these breaches.

Real-World Cases (Technical Focus)



Target Data Breach

2013

HVAC vendor credentials compromised;

Forensic Finding: NetFlow review showed unusual outbound traffic to Eastern Europe.

Lesson: Segmentation & anomaly detection reduce damage.



WannaCry Ransomware

2017

Exploited SMBv1 (EternalBlue);

Forensic Finding : worm-like scanning across port 445; kill-switch domain found via DNS analysis. The infamous “kill switch domain” was identified through DNS traffic analysis.

Lesson: Patch management is critical.



Davidoff's Packet Captures

2013

Packet captures exposed IRC-based C2 channels via string searches in pcaps.

Forensic Finding: Investigators used string searches in pcaps to reveal hidden attacker communication.

Lesson: Benign-looking traffic can reveal infrastructure.



Sony Pictures Hack

2014

~100TB exfiltrated; hardcoded C2 IPs in malware; months of data staging before the destructive attack.

Forensic Finding: Forensic review revealed hardcoded IP addresses in malware communicating with C2 servers. Network logs confirmed months of data staging before the final destructive attack.

Lesson: Early outbound monitoring is essential.

Chapter Summary

- Network security layers (perimeter, host, application, monitoring) generate the evidence forensic investigators rely on.
- Digital evidence is fragile: follow the Order of Volatility, verify integrity with hashing, and maintain an unbroken chain of custody.
- A structured methodology — prepare, collect, analyze, respond, document — underpins effective investigations.
- High-profile breaches (Target, WannaCry, Sony) show that network monitoring and timely patching limit damage.